

Stakeholder Engagement

The two selected stakeholder groups are Community members (African language speakers & community representatives), engaged through Distributed Dialogue, and Institutions (African linguists & language researchers), engaged through Participatory Design.

1. Community members - (African Language Speakers & Community Representatives)

African language speakers and community representatives are primarily concerned with whether SuperBPE represents their languages accurately, respectfully, and without treating them as secondary to higher-resource languages. They may value cultural preservation, linguistic identity, inclusion, and continued use of their languages in digital environments. They may also be concerned about their language being reduced to statistical patterns that fail to capture important expressions, spelling conventions, diacritics, or culturally meaningful forms. They would benefit from AI technologies that make their languages more visible and usable, but may be concerned that poor representation could reinforce existing inequalities rather than reduce them. Their participation can therefore help establish what “good representation” means from the community’s perspective.

Potential benefits

- Greater representation of African languages in AI.
- Support for continued digital use of local languages.
- Better representation of culturally meaningful expressions.
- Greater community involvement in AI development.
- Identification of linguistic problems that technical evaluation might miss.
- Increased trust in African-language technology.

Potential harms

- Misrepresentation of language or cultural expressions.
- Overrepresentation of languages with larger datasets.
- Underrepresentation of very-low-resource languages.
- Incorrect normalization or handling of spelling/diacritics.
- Community knowledge being used without meaningful participation or acknowledgement.
- Creating the appearance of inclusion while underlying language-model performance remains poor.

Engagement Method: Distributed Dialogue

Based on the Action Catalogue, Distributed Dialogue would be used to engage African language speakers and community representatives because SuperBPE concerns multiple languages and communities rather than a single homogeneous population. A centralized workshop could unintentionally give greater visibility to participants who live near research institutions, have technical knowledge, or speak languages with stronger digital representation. A distributed approach

provides an opportunity to gather perspectives from people in different linguistic and geographic contexts.

The purpose of the dialogue would not be to ask participants to design the tokenizer's algorithm. Instead, participants would help define what successful and respectful representation of their languages should look like. A participant does not need to understand BPE or neural networks to identify that a particular word, expression, spelling convention, or culturally meaningful phrase has been represented poorly.

The engagement could involve presenting participants with carefully selected examples of how the tokenizer segments words and short expressions. Participants could identify cases they consider unnatural, confusing, or potentially misleading. They could also provide examples of language features that they believe should be preserved. Feedback could be collected through accessible questionnaires, small online discussions, community groups, or structured conversations with representatives where direct participation is difficult.

A key principle would be two-way communication. Participants should be told what the project is trying to achieve, what data is being considered, and what tokenizer outputs are being observed. They should also be told how their feedback will influence the project. This avoids treating stakeholder engagement as simply collecting data from communities.

The dialogue should deliberately include perspectives from both relatively well-resourced and under-resourced languages. This is important because a tokenizer optimized primarily according to corpus frequency could favour languages with larger datasets. Community participation can therefore reveal whether aggregate technical improvements hide poor representation for smaller language communities.

The engagement would include a feedback loop. After incorporating community feedback, the team could return with revised examples and ask whether the identified problems have improved. Where suggestions cannot be implemented, the project should document why. This creates accountability without promising that every community request can be incorporated.

2. Institutions - (African Linguists & Language Researchers)

African linguists and language researchers are primarily concerned with linguistic accuracy, preservation of language-specific features, and reliable evaluation across different African languages. They would want SuperBPE to account for differences in morphology, orthography, diacritics, word formation, and language variation rather than assuming that one tokenization strategy is equally appropriate for every language. They are also likely to value transparent methodology, reproducible experiments, and appropriate datasets. Their main concern may be that an algorithm optimized for compression or token frequency produces technically efficient tokens while ignoring important linguistic structures. Their expertise can therefore help distinguish statistical efficiency from meaningful linguistic representation.

Potential benefits

- More linguistically informed tokenizer design.
- Better representation of morphology and orthographic variation.
- Stronger evaluation methodology.
- Identification of language-specific weaknesses.
- Improved dataset selection and normalization decisions.
- Increased research value and credibility.

Potential harms

- Linguistic expertise being consulted too late to influence design.
- Incorrect assumptions being made about African languages.
- A single tokenizer being treated as equally optimal for every language.
- Language-specific characteristics being lost during normalization.
- Evaluation focusing on aggregate performance and hiding poor performance for individual languages.

Engagement Method: Participatory Design

Participatory Design is particularly appropriate for African linguists and language researchers because they possess domain expertise that can directly inform the design of SuperBPE. Rather than asking linguists to evaluate a finished tokenizer at the end of development, their involvement would commence during the design and evaluation process so that their knowledge can influence technical decisions before they become difficult to change.

The engagement would begin by presenting the project's objectives, proposed language-selection framework, dataset sources, preprocessing decisions, and intended two-stage SuperBPE approach. Linguists could review these assumptions and identify areas where the technical design might conflict with the characteristics of particular languages.

Participants would then work with the technical team on representative examples. Instead of evaluating only a final compression score or vocabulary size, they could examine how words and expressions from different languages are segmented. This

could reveal cases where the tokenizer achieves good statistical compression but produces linguistically undesirable segmentation.

Their feedback could also inform dataset construction. Linguists could help identify appropriate sources, problematic texts, spelling variations, dialectal differences, and potentially misleading data. This is particularly important for low-resource languages, where a small dataset may not adequately represent the language community.

Participatory Design would also create a structured way to negotiate disagreements. For example, a technically efficient merge may produce good benchmark results but be inappropriate for a particular language. Rather than assuming that one metric should determine the decision, the team could document the trade-off and decide whether additional evaluation is necessary.

The process should include several cycles: design → prototype → linguistic review → modification → re-evaluation. This makes stakeholder input actionable rather than symbolic. The technical team would remain responsible for implementation, while linguistic experts would contribute domain knowledge and help establish appropriate evaluation criteria.

For SuperBPE, participatory design could therefore help bridge the gap between statistical tokenization efficiency and meaningful linguistic representation. It would make the tokenizer more defensible scientifically while reducing the risk of embedding incorrect assumptions about African languages into the system.

How Engagement Influences Dataset and Model Design

The engagement would directly influence both the dataset and tokenizer design. Community feedback could identify culturally important expressions, spelling variations, and language-specific cases that should be represented in evaluation data. Linguists could help review language selection, data sources, normalization, orthographic variation, and evaluation criteria. Their input could also lead to language-specific test sets rather than relying only on aggregate tokenizer performance. If stakeholder feedback reveals that high-frequency merges disadvantage particular languages, the SuperBPE training strategy or vocabulary allocation could be adjusted. Feedback would therefore become part of an iterative cycle of data selection → tokenizer training → stakeholder evaluation → revision, rather than being collected only after development.